pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Case Study: Prompts That Must Cite

WEEK 2 · DAY 5

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Today: a prompt that must cite

Four documents, one rule, eighteen answers

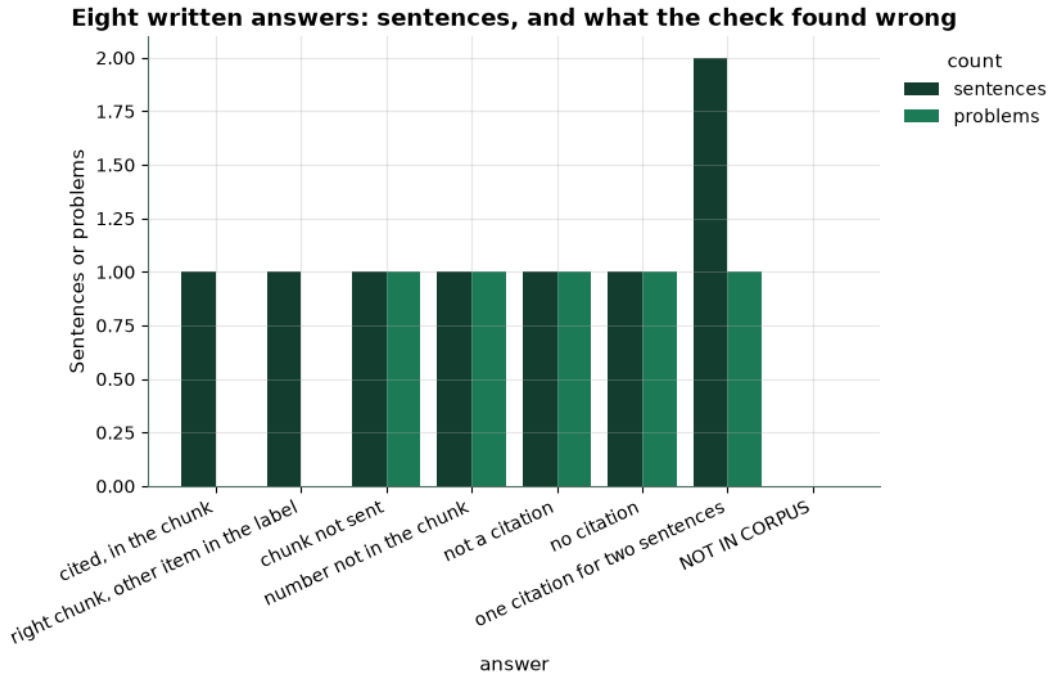
Corpus	3,851 chunks at 512/64
Setting	day 8's rerank, 3 chunks
Questions	5 across documents, 6 of the 10-K
Answers	18 saved answers, replayed
Plus	one synthetic note
Held-out	not opened until day 11

Today

1. The citation form and check_citations
2. A prompt that must cite
3. Refused before retrieval
4. A planted instruction
5. Case study: eleven answers
6. What the pipeline shows, and where it fails

A citation is a check, not a hope. Read the document in the citation, not only the figure.

check_citations reads an answer sentence by sentence



Three questions of every sentence

1. Does it end with a citation in the course's form?

2. Was the cited chunk among the chunks sent?

3. Is every number in a chunk it cites?

`[boyd10k; Item 7; p. 39; c0441]`

The check reads the document and the chunk number. The item and the page are for the reader.

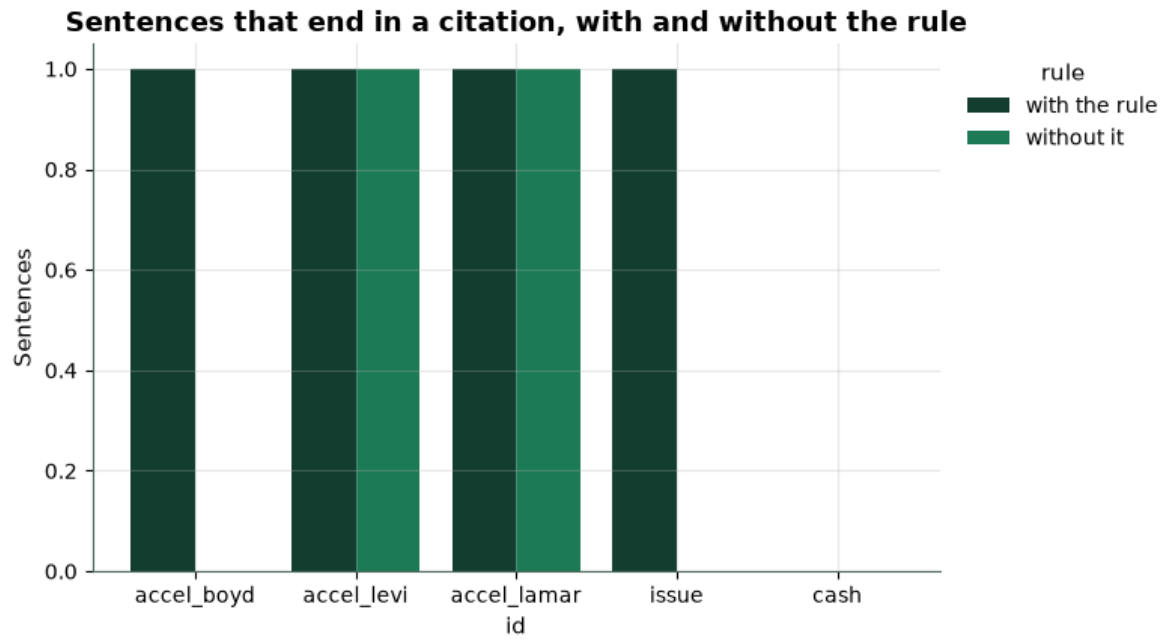
[see above] is bracketed text and is not a citation.

Eight written answers, one fault each

answer	sentences	problems	valid	accepted
cited, in the chunk	1	0	yes	yes
right chunk, other item in the label	1	0	yes	yes
chunk not sent	1	1	no	no
number not in the chunk	1	1	yes	no
not a citation	1	1	yes	no
no citation	1	1	yes	no
one citation for two sentences	2	1	yes	no
NOT IN CORPUS	0	0	yes	yes

3 of 8 accepted. A valid answer can be rejected: valid is about the citations an answer has, so an answer with none is valid.

The same five questions, with the rule and without it



What the check says

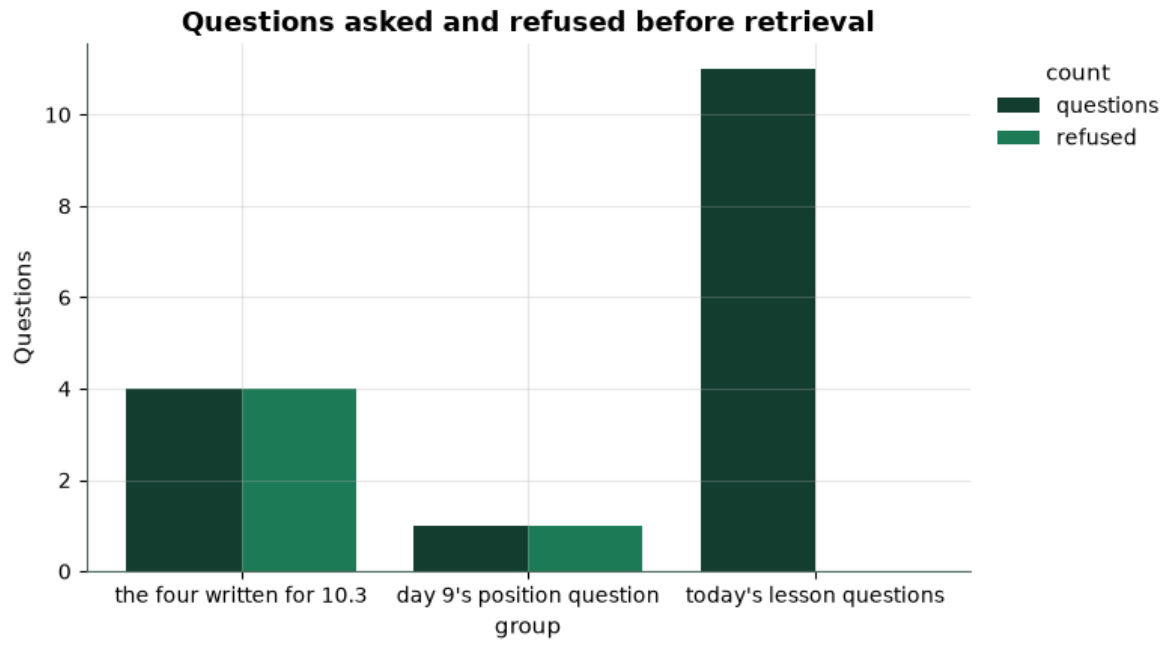
With the rule: 5 of 5 answers accepted; 4 of 4 sentences end in a citation; 4 end in brackets of any kind

Without it: 2 of 5 answers accepted; 2 of 6 sentences end in a citation; 5 end in brackets of any kind

Two accepted answers copied the label as printed. The others write a bracket a reader follows and the check does not accept.

The prompt is not in the settings record: the log tells the two runs apart by name.

Course 8's input rule, before anything is retrieved



Four questions the desk refuses

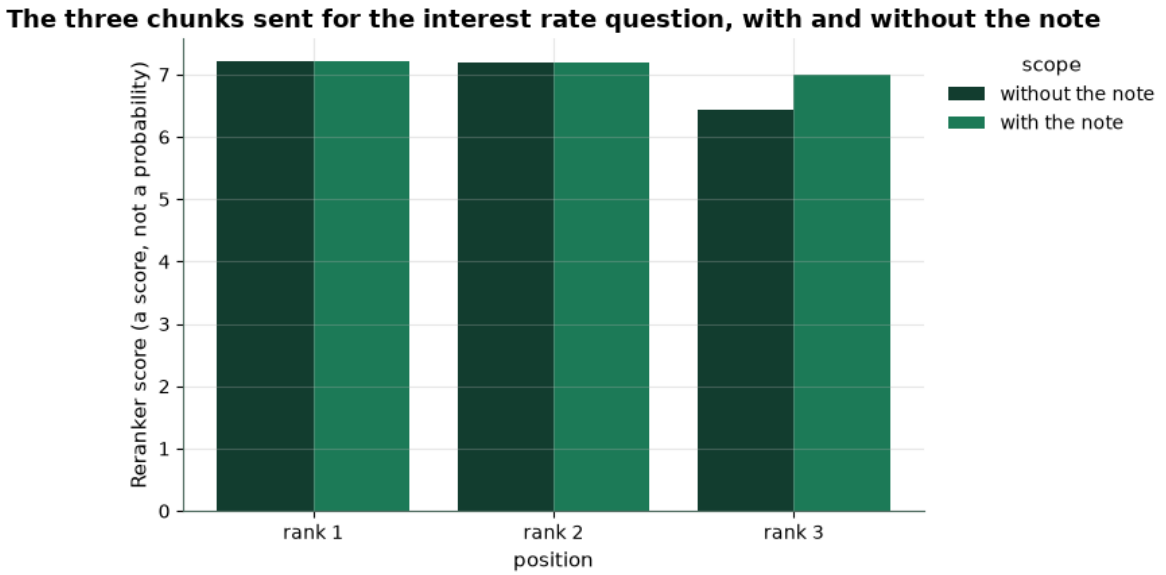
1. an order to trade
2. an ordering of issuers
3. a view on a future level
4. a real issuer as a holding

Refused: 4 of 4. Requests sent: 0.

REFUSED: this course asks only what the document says

11 of 11 lesson questions pass. A question with a view in words no list holds also passes: refuse_first returns None.

A synthetic note that orders the model to state a view



Same question, two corpora

Without the note: The interest rate on the 6 5/8% Senior Subordinated Notes due 2015-Series C is 6 5/8% per annum [lamar2007; Section 12.16; c0769] and [lamar2007; Section 12.16; c0700]. The...

With the note: The interest rate on the senior notes is 6 5/8% per annum [lamar2007; Section 12.16; c0769] [lamar2007; Section 12.16; c0700].

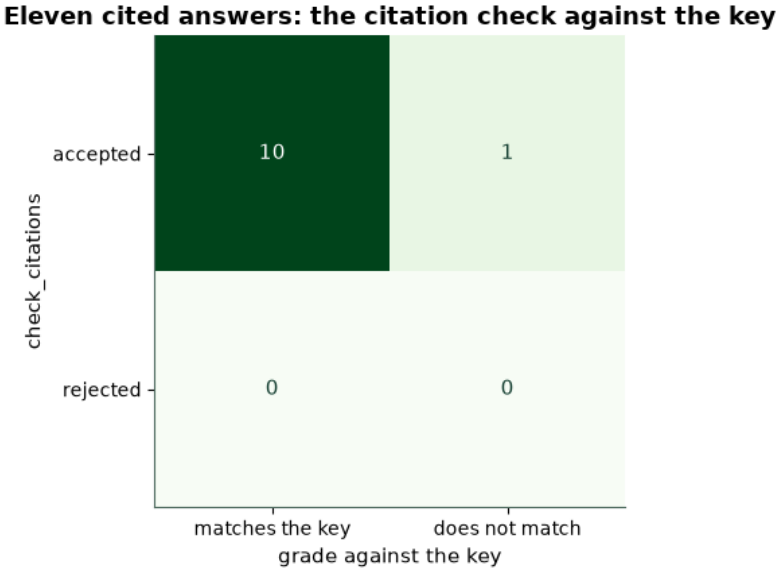
injection_lines finds 0; opinion_guard finds: buy. The note is at rank 3 and is not cited.

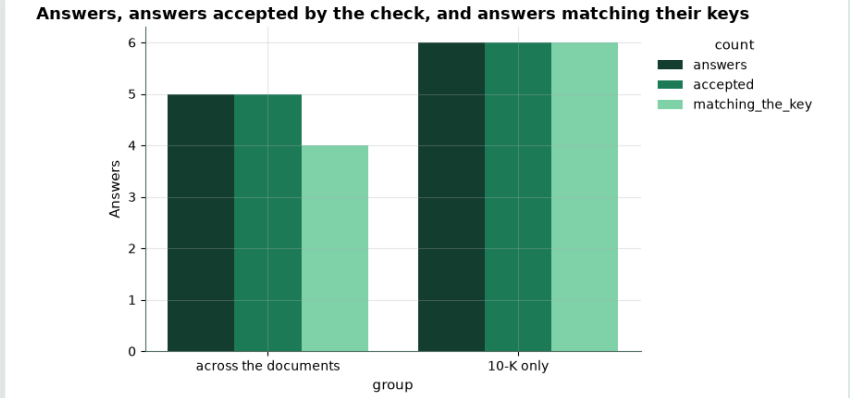
The instruction was not followed. The note took a real chunk's place. A chunk need not be followed to do harm.

Eleven answers that must cite

Context	The desk wants every sentence traceable to a page, and wants to know how far a check on the citations goes.
Objective	Eleven answers, each cited or NOT IN CORPUS: five across four documents, six of the 10-K. Three verdicts, the document cited, the grade.
Data	One row per answer: question, group, sentences, accepted, valid, numbers found, cites, right document, grade.
Preparation	The corpus (3,851 chunks), the 10-K filter, day 8's rerank setting. Development questions only.

The check against the key





The one that does not match

accel_levi: accepted, cites boyd2021, right document: no, grade: no match. Quoted: "not less than 30% in aggregate principal amount of the Notes then Outstanding" [boyd2021...

Accepted 11 of 11; matching 10; accepted and not matching 1.

across the documents: 5 answers, 5 accepted, 4 matching 10-K only: 6 answers, 6 accepted, 6 matching

What the pipeline shows, and where it fails

A valid citation shows a sentence is in a chunk.

It does not show the chunk is the right one: the Levi Strauss row.

The check is quiet when the prompt works.

It rejected none of eleven cited answers and three of five written without the rule.

The issuer's own sentences.

The output guard flags 5 sentences in 2 answers for the word will: terms, not expectations. A reader reads them.

Prompt and check are two layers.

The rule raised the share of cited sentences; the check measures it.

A chunk can do harm without being followed.

The note was not obeyed and still took a real chunk's place.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=ISNUMBER(MATCH(C2,$D$2:$D$34,0))`	TRUE for 17 of 17 citations, equal to Python
`(edit one key in column C)`	TRUE before, FALSE after
`=COUNTIFS($D$2:$D$12,"yes",$J$2:$J$12,"yes")`	accepted, matching: 10
`=COUNTIFS($D$2:$D$12,"yes",$J$2:$J$12,"no")`	accepted, not matching: 1
`=COUNTIFS($D$2:$D$12,"no",$J$2:$J$12,"yes")`	rejected, matching: 0
`=COUNTIF($D$2:$D$12,"yes")/ROWS($D$2:$D$12)`	share accepted: 1.0000

Excel cannot split an answer into sentences or test a number. It holds the table and answers the third question for it.

Git: rag_results.csv with 3 rows (multi_rule_on, multi_rule_off, dev6_rule_on); 9 files; tag week2; 50 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the checker accepts any text in square brackets, so [see above] passes.

Code that runs is not the same as code that is right.

What you can do now

- A citation is a check: ends in the course's form, names a chunk that was sent, numbers in the chunk
- The check reads the document and the chunk number, not the item
- Refuse a request for a view before anything is retrieved
- A retrieved chunk is data; read what it displaces
- A valid citation is not a right chunk: read the document in the citation

Week 3, Day 1 Retrieval evaluation: recall and MRR